

Surface Areas and Volumes

Chapter 12 - Hinglish Notes, Formulas, Examples + Solutions

Ye Chapter 12 (Surface Areas and Volumes) ka complete guide hai. Pehle red-page ke core basics (SA vs Volume, CSA vs TSA, pi, slant height, shapes ke naam, units), fir chapter ka main content - har solid ke CSA/TSA/Volume formulas, combination of solids aur frustum. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Surface Area vs Volume

EXAM * Exam me CONSTANTLY aata hai

- Surface Area = bahar ki area (paint karne wali jagah), unit cm^2 .
- Volume = andar ki jagah (paani aa sake jitni), unit cm^3 .

2. CSA vs TSA

EXAM * Exam me CONSTANTLY aata hai

- CSA = Curved / Side Surface (top-bottom chhod ke).
- TSA = Total Surface Area (saari surfaces milake).

3. pi (PI)

- $\pi = 22/7$ (radius 7 ka multiple ho to) ya 3.14.

4. Square aur Cube

EXAM * Exam me CONSTANTLY aata hai

- $a^2 = a \times a$ (area me).
- $a^3 = a \times a \times a$ (volume me).

5. Square Root (slant height ke liye)

EXAM * Exam me CONSTANTLY aata hai

- Cone me slant height: $l = \sqrt{r^2 + h^2}$.
- Ye Pythagoras se aata hai.

6. Basic Shapes ke Naam

- Cube, Cuboid, Cylinder (belan), Cone (shanku),
- Sphere (gol), Hemisphere (aadha sphere).

7. Units

- Area $\rightarrow \text{cm}^2$, Volume $\rightarrow \text{cm}^3$. Mat bhoolna!

PART B - Chapter 12 ka Main Content

8. Cuboid aur Cube

EXAM * Exam me CONSTANTLY aata hai

CUBOID (l, b, h):

$$\text{TSA} = 2(lb + bh + hl), \quad \text{Volume} = l \times b \times h$$

CUBE (side a):

$$\text{TSA} = 6a^2, \quad \text{Volume} = a^3$$

9. Cylinder

EXAM * Exam me CONSTANTLY aata hai

$$\text{CSA} = 2\pi r h$$

$$\text{TSA} = 2\pi r (r + h)$$

$$\text{Volume} = \pi r^2 h$$

10. Cone

EXAM * Exam me CONSTANTLY aata hai

$$\text{Slant height } l = \sqrt{r^2 + h^2}$$

$$\text{CSA} = \pi r l$$

$$\text{TSA} = \pi r (l + r)$$

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$

11. Sphere aur Hemisphere

EXAM * Exam me CONSTANTLY aata hai

$$\text{SPHERE: Surface} = 4\pi r^2, \quad \text{Volume} = \frac{4}{3} \pi r^3$$

$$\text{HEMISPHERE: CSA} = 2\pi r^2, \quad \text{TSA} = 3\pi r^2$$

$$\text{Volume} = \frac{2}{3} \pi r^3$$

12. Combination of Solids

EXAM * Exam me CONSTANTLY aata hai

- Combined shape ka surface = bahar dikhne wali surfaces ka jod.
- Combined volume = alag-alag solids ke volumes ka jod.
- Example: ice-cream cone = cone + hemisphere.

13. Frustum of a Cone

EXAM * Exam me CONSTANTLY aata hai

Cone ka upar ka hissa kaat do to frustum banta hai (R, r, h):

Volume = $(1/3) \pi h (R^2 + r^2 + Rr)$
CSA = $\pi (R + r) l$, $l = \sqrt{h^2 + (R-r)^2}$
TSA = CSA + $\pi R^2 + \pi r^2$

Quick Revision

Cube: TSA = $6a^2$, V = a^3 . Cuboid: V = lbh .
Cylinder: CSA = $2\pi r h$, V = $\pi r^2 h$.
Cone: CSA = $\pi r l$, V = $(1/3)\pi r^2 h$, $l = \sqrt{r^2 + h^2}$.
Sphere: $4\pi r^2$, V = $(4/3)\pi r^3$. Hemisphere V = $(2/3)\pi r^3$.
Combination: surfaces/volumes jodo. Area cm^2 , Volume cm^3 .

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1: Cube ka side 5 cm. Volume aur TSA?

- Step: Volume = $a^3 = 5 \times 5 \times 5 = 125$.
- Step: TSA = $6a^2 = 6 \times 25 = 150$.

Answer: Volume = 125 cm^3 , TSA = 150 cm^2

Example 2: Cuboid l=4, b=3, h=2. Volume?

- Step: Volume = $l \times b \times h = 4 \times 3 \times 2$.

Answer: 24 cm^3

Example 3: Cylinder r=7, h=10. Volume? ($\pi=22/7$)

- Step: Volume = $\pi r^2 h = (22/7) \times 49 \times 10$.
- Step: = $22 \times 7 \times 10$.

Answer: 1540 cm^3

Example 4: Cylinder r=7, h=10. CSA? ($\pi=22/7$)

- Step: CSA = $2\pi r h = 2 \times (22/7) \times 7 \times 10$.
- Step: = $2 \times 22 \times 10$.

Answer: 440 cm^2

Example 5: Cone r=3, h=4. Slant height l?

- Step: $l = \sqrt{r^2 + h^2} = \sqrt{9 + 16}$.
- Step: = $\sqrt{25}$.

Answer: $l = 5 \text{ cm}$

Example 6: Cone r=3, h=4. Volume? ($\pi=3.14$)

- Step: V = $(1/3) \pi r^2 h = (1/3) \times 3.14 \times 9 \times 4$.
- Step: = $(1/3) \times 113.04$.

Answer: 37.68 cm^3

Example 7: Sphere r=7. Surface area? ($\pi=22/7$)

- Step: SA = $4\pi r^2 = 4 \times (22/7) \times 49$.
- Step: = $4 \times 22 \times 7$.

Answer: 616 cm^2

Example 8: Sphere r=3. Volume? ($\pi=3.14$)

- Step: $V = \frac{4}{3} \pi r^3 = \frac{4}{3} \times 3.14 \times 27$.
- Step: $= \frac{4}{3} \times 84.78$.

Answer: 113.04 cm^3

Example 9: Hemisphere $r=7$. TSA? ($\pi=22/7$)

- Step: $TSA = 3 \pi r^2 = 3 \times \frac{22}{7} \times 49$.
- Step: $= 3 \times 22 \times 7$.

Answer: 462 cm^2

Example 10: Cone ($r=3, h=4$) ke upar hemisphere ($r=3$). Exposed surface?

- Step: Cone CSA $= \pi r l = \pi \times 3 \times 5 = 15 \pi$.
- Step: Hemisphere CSA $= 2 \pi r^2 = 2 \pi \times 9 = 18 \pi$.
- Step: Jodo: $15 \pi + 18 \pi$.

Answer: $33 \pi \text{ cm}^2$

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1. Cube ka volume formula kya hai?
- Q2. Cube ka TSA formula kya hai?
- Q3. Cuboid ka volume formula kya hai?
- Q4. Cylinder ka volume formula kya hai?
- Q5. Cylinder ka CSA formula kya hai?
- Q6. Cone ka volume formula kya hai?
- Q7. Cone ki slant height ka formula kya hai?
- Q8. Cone ka CSA formula kya hai?
- Q9. Sphere ka volume formula kya hai?
- Q10. Sphere ka surface area formula kya hai?
- Q11. Hemisphere ka volume formula kya hai?
- Q12. Hemisphere ka TSA formula kya hai?
- Q13. CSA aur TSA me kya antar hai?
- Q14. Area aur volume ke units kya hote hain?
- Q15. Cube side=4 -> volume?
- Q16. Cylinder $r=7, h=5$ -> volume? ($\pi=22/7$)
- Q17. Cone $r=6, h=8$ -> slant height?
- Q18. Sphere $r=7$ -> surface area? ($\pi=22/7$)
- Q19. Frustum ka volume formula kya hai?
- Q20. Combination of solids me total volume kaise nikalte hain?

Step-by-Step Solutions

Q1. Cube ka volume?

- Step: side ka cube.

Answer: a^3

Q2. Cube ka TSA?

- Step: 6 faces, har ek a^2 .

Answer: $6a^2$

Q3. Cuboid ka volume?

- Step: $l \times b \times h$.

Answer: $l b h$

Q4. Cylinder ka volume?

- Step: base area $\times$ height.

Answer: $\pi r^2 h$

Q5. Cylinder ka CSA?

- Step: curved surface.

Answer: $2 \pi r h$

Q6. Cone ka volume?

- Step: cylinder ka $\frac{1}{3}$.

Answer: $(\frac{1}{3}) \pi r^2 h$

Q7. Cone slant height?

- Step: Pythagoras.

Answer: $l = \sqrt{r^2 + h^2}$

Q8. Cone ka CSA?

- Step: curved surface.

Answer: $\pi r l$

Q9. Sphere ka volume?

- Step: standard.

Answer: $(\frac{4}{3}) \pi r^3$

Q10. Sphere ka SA?

- Step: standard.

Answer: $4 \pi r^2$

Q11. Hemisphere ka volume?

- Step: sphere ka aadha.

Answer: $(\frac{2}{3}) \pi r^3$

Q12. Hemisphere ka TSA?

- Step: curved + flat circle.

Answer: $3 \pi r^2$

Q13. CSA vs TSA?

- Step: curved only vs sab milake.

Answer: CSA = side only; TSA = total

Q14. Area aur volume units?

- Step: 2D vs 3D.

Answer: cm^2 aur cm^3

Q15. Cube side=4 $\rightarrow$ volume?

- Step: $4^3 = 64$.

Answer: 64 cm^3

Q16. Cylinder $r=7, h=5 \rightarrow$ volume?

- Step: $(\frac{22}{7}) \times 49 \times 5 = 22 \times 7 \times 5$.

Answer: 770 cm^3

Q17. Cone $r=6, h=8 \rightarrow$ slant?

- Step: $\sqrt{36+64} = \sqrt{100}$.

Answer: 10

Q18. Sphere $r=7 \rightarrow$ SA?

- Step: $4 \times (22/7) \times 49 = 4 \times 22 \times 7$.

Answer: 616 cm^2

Q19. Frustum ka volume?

- Step: standard formula.

Answer: $(1/3) \pi h (R^2 + r^2 + R r)$

Q20. Combination total volume?

- Step: alag solids ke volume jodo.

Answer: Volumes ka jod